

Galaxy

SCIENCE FICTION

JULY 1958

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BULLET WITH
HIS NAME

By

**FRITZ
LEIBER**

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THE BACK
OF
OUR HEADS

By

**STEPHEN
BARR**

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IGY
ROUNDUP

By

**WILLY
LEY**

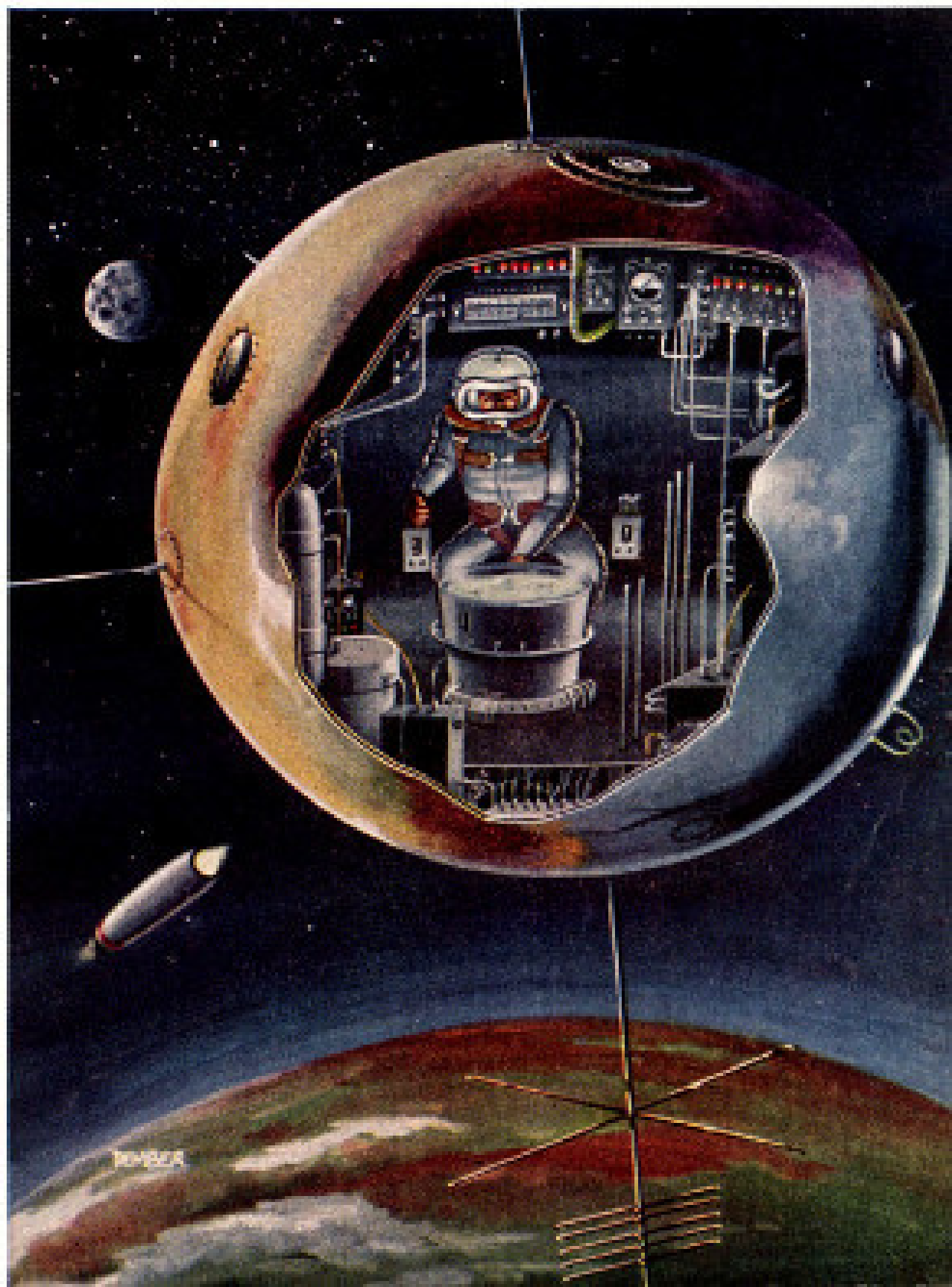
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INNOCENT
AT LARGE

By

**POUL & KAREN
ANDERSON**

And Other Stories



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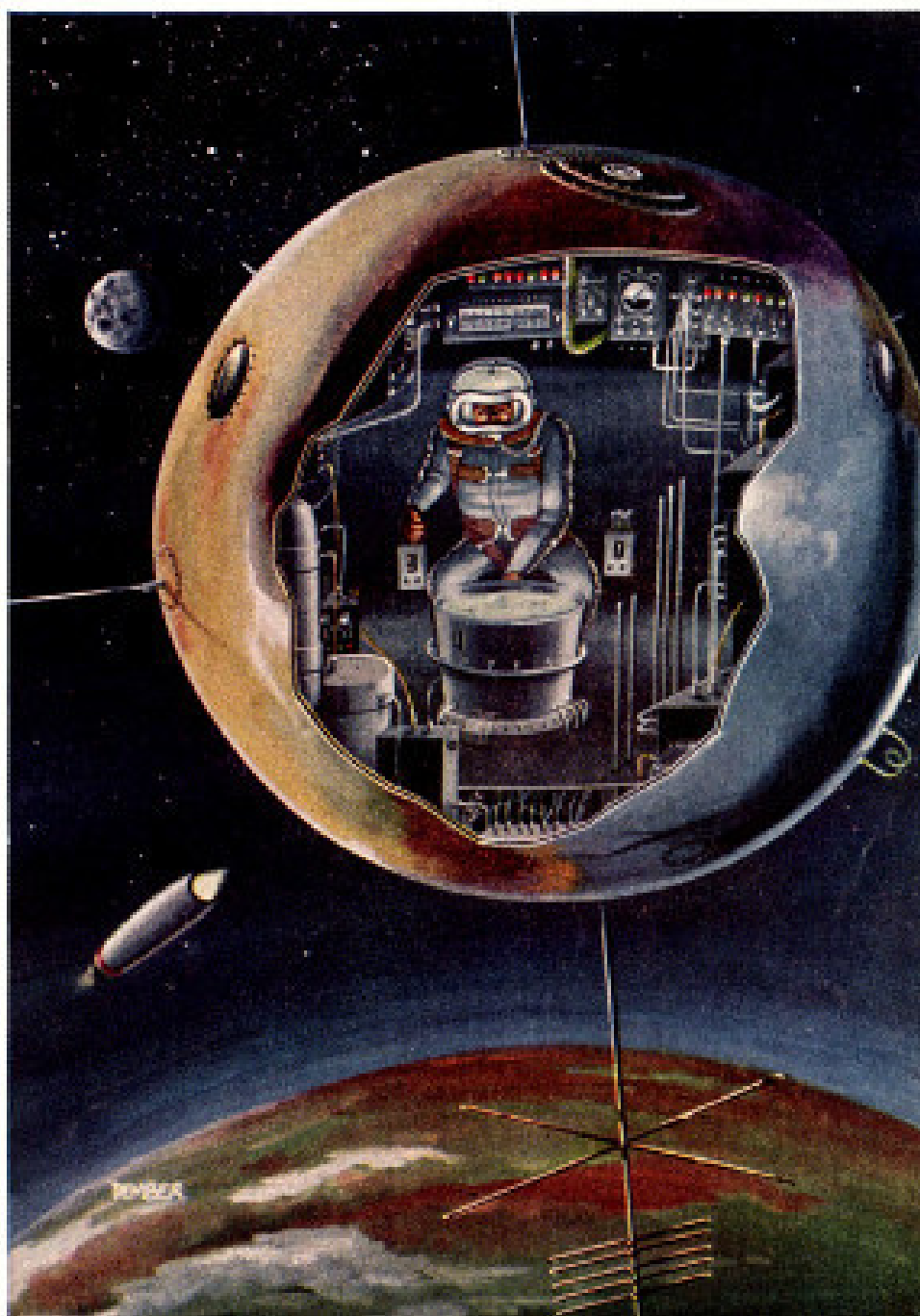
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*** START OF THE PROJECT GUTENBERG EBOOK THE BACK OF
OUR HEADS ***

The Back of Our Heads

By STEPHEN BARR

Illustrated by DILLON

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***She traveled from life to death and back again
like a commuter on the 5:15 ... except each trip
brought her nearer the beginning of the line!***

In reading this report, it must be borne in mind that when the word "they" is used, it does not refer necessarily to separate entities as individuals.

It is possible that a closer analogy would be the cells of an organism—which, in a sense, we ourselves become when we are in a pack or forming a mob.

On the other hand, that particular cell or entity which this report deals with exhibited at all times marked individuality—even eccentricity—and will hereinafter be referred to as "she." This is because "she" invariably assumed a female form when visiting us, and because she furthermore gave every indication of that type of mind and point of view which is generally met with in the more noticeable, effective or contentious members of that sex.

As she put it herself during the hearing, she was always in hot water.



The four teen-agers—one girl, three boys—weren't allowed in the bar, so they went down the street to a joint where there were a soda fountain, booths and a jukebox. They sat in a booth and a waitress came to take the orders: three hot dogs and three cokes.

"What about you, dear?"

"Just a glass of water." The waitress started to leave. "No, wait—gimme a white on rye, too."

The waitress left, then came back again. "What was that you wanted, dear? Some kind of rye-bread sandwich?"

"Changed my mind. Make it a buttered pecan, but tell 'em to go easy on the butter. And I don't want no French dressing. Make it on whole wheat."

The waitress looked uncertain. "You mean a *nut* sandwich?"

"Yeah, only malted. With lettuce and chocolate sprinkles."

"Who you kiddin'?" the waitress said, and turned to go.

"No, hold it. Tell Joe to please scramble them on both sides."

"What you *talkin'* about?" the waitress said. "We ain't got no one here called Joe."

"So okay, Joseph, then. Tell him just a boiled egg sunny side up."

The waitress left, frowning.

"Our Miss Framis," one of the boys said, meaning the girl, and the others smiled. They looked as though they were sneering at the same time and hoped they would be taken for juvenile delinquents.

There were two very odd-looking men in the booth opposite and they were listening to the conversation. Their oddness lay in an atmosphere rather than in any physical abnormality. The girl noticed them and nudged one of the boys.

The three boys looked at the men resentfully and one of them said something under his breath, but the girl said, "Button it." Then she asked the men opposite, "Lookin' for someone, mister?"

The two men looked away, and this made the boys feel brave. One of them said, "Let's give 'em the works."

"No, leave it to me." The girl got up and went across to the two men. "Me and my friends was wondering. Maybe you gentlemen would like to come to a trake in the gort later?"

The three boys snickered and the men looked up at the girl and waited with blank faces.

"Or maybe you'd rather we put on a hanse for you?" she said.

"No, sit down," one of the men—the bigger one—said, and moved back to make room for her. She glanced at him with surprise for a moment and sat down next to him.

One of the boys started to get up when he saw this, but the others pulled him down again.

"What did you say to us just now?" the big man asked. "It was too small in here."

She shook her head and frowned. "Why, that was just ... I said did you want for us to put on a hanse, is all." She had a rather feeble grin.

"Yes," the big man said. "We do."

She glanced back at her friends nervously, and then at the man again. "I don't get you," she said.

"Neither do we," the smaller man said.

The boys across the room were listening quietly and then one of them said, "Go on, tell 'em, Miss Framis."

"We just want you to quint," the big man said, "and won't thursday on it."

She stared at him without expression and got up slowly. She went over to her friends. "Let's get out of here," she said.

She was shivering.

Q. You say you object to this line of questioning?

A. (She) No, I just don't like being spied on. And it made the kids ... mad. They wrecked the car and that meant starting all over again.

Q. The car?

A. Yes, their hot-rod. When we got outside, they acted the way teen-agers do and went too fast. They were sore at those spies—they took it out on the car, so it went off the road. It turned over three times and we were all killed.

Q. They were not spies. They were acting on their own.

A. *I* didn't know that. I just knew something was funny. Anyway, how can you say that? They're a waste. And it would have been part of you, just as I am. It was more of a waste if I hadn't been split. The other part was only about eleven years old and I had to wait another six years to—

Q. It is your own fault if you were split. You cannot blame us. This has happened before—you have aimed badly and arrived wrong. Don't forget about the help.

A. Well, in this case it's a lucky thing I did; otherwise the whole thing would have been wasted. And the kelp—that was dreadfully dull. I wanted to try a really primitive form, but not *that* primitive. Then I got washed up and it led to the cat. After they got the iodine out of the kelp, I was suddenly a cat.

Q. This has not been reported.

A. I'm reporting it now. It wasn't dull in the least, but they were very superstitious about cats in those days, and they decided I was possessed.

Q. They saw through you?

A. Oh, yes. People usually do.

Q. You couldn't have been very successful if they saw through you.

A. It doesn't make any difference if they see through you. The important thing is to see through them.

Q. But you were a cat.

A. Cats are in a very good position to see through people. I think they sensed that. Anyway, I was ... done away with.

Q. Burned again?

A. Yes.

Q. Seems to be a habit of yours. What happens? How does it feel?

A. I cannot explain it to you, but I know what to do. It's not *my* habit—it's one of theirs, but it's dying out in most places now. And there was a time when it would never have occurred to them. They were too frightened of it.

Q. Frightened of what?

A. Of fire. It was very new then....

The hunters came back to the cave at dusk, and one of them went to the fire that was kept going constantly in front of the opening. He took a dry branch and held it in the fire until the end caught. Then he held it up. "If we take this, we can hunt in the dark," he said. "And when it is nearly eaten by the fire, we can take another branch and start it again. That way we do not need the moon."

"That way we can hunt until we are tired," said the other.

"That way we can kill twice as much game," said the first.

"There is so much game in the cave now," a young woman said, "that it is beginning to smell."

The older hunter glanced at her apprehensively; she made him feel foolish, always finding fault with his plans. "Perhaps so," he said. "But at other times we starve."

"Besides," she said, "if you take the fire with you to see where you are going and to see the game, the game will see you."

The hunters looked at one another and shrugged. The woman went into the cave and returned with an earthenware pot. There were pieces of raw meat and some water in it and she put it on the fire, propping it in position with three stones. The second hunter looked at the pot curiously. He was a younger brother from the other side of the valley, where he lived with his mates. He pointed at the pot and looked inquiringly at the older brother.

"She made it out of mud," the older brother said.

"Why doesn't it fall apart with the water in it?"

"I put it into the fire first, for a long time," the young woman said. "A very big fire. The mud gets red—and then it gets hard so it won't melt when the water is in it."

The younger man looked surprised. "Magic?"

"Yes," said the other man.

"Nonsense," said the woman. She went back to the cave and the young man put the end of his spear into the fire and tried to scrape the side of the pot with the flint head, but the flint was cold and it cracked. He pulled it back and was looking angrily at it when she came out again and sat on the ground. She had an armful of roots which she began to scrape with a sharp stone. "The spearhead is made of the wrong sort of stone," she said, without looking up. "That is why it broke in the fire."

"It's made of the *right* kind!" the young man shouted. "All spearheads are made of that kind! They always have been and they always will be! How did *you* know it broke in the fire? You weren't looking."

"I heard it make the sound it makes when the fire breaks it."

The young man glowered and pushed his under lip out. "This kind of stone was put in the cave for us to make knives and spears. And it makes a very sharp edge when you know how to form it."

"No sharper than this knife," she said, holding up the stone in her hand. "This doesn't break so easily."

The young man took it and examined it carefully. "How do you strike it to make it this shape?" he said, and then, grudgingly, "It is very smooth—a very good shape."

"You don't strike it," she said, taking it back and going on scraping the roots. "You rub it on another stone—first on the kind that has the bright sparkles in it, and then under water on the flat gray kind. It's much better than your knives and the fire doesn't break it so easily."

She finished with the roots and put them in the pot with the meat.

"Where do you find such big roots?" the young man asked his brother.

"Over there," the brother said, pointing to a patch of earth nearby. "*She* finds them there."

"I don't find them," she said. "I put them there in the first place."

"You mean you store them in the earth?" the young man said.

"No. I put the tops in the ground—the blue and yellow flowers—and next warm season I dig and there are the new roots. You have to put water on the earth when it gets dry. Also you have to pull up all the small plants that grow there among them. It's very hard work."

"More magic," said the young man.

"It's not magic!" she said. "You are stupid. Haven't you noticed that when you leave an acorn on the ground, it breaks open and a finger goes down into the earth? And then, after the next rains, it makes little leaves—and if you leave it alone, it grows and in time becomes a young tree?"

"Everyone knows that," the young man said disdainfully.

"Well, this is the same."

"Yes, but what makes the roots so big? I never saw any like these."

"That's because I only take the flowers from the plants that have the biggest roots. And if any of the new roots are little, I throw away the flowers from them. Far away."

"What do you do with the little roots?"

"Eat them."

"I don't understand. If you eat the little roots, why don't you get little roots?"

"You are being foolish again!" the young woman said. "A tall man has tall sons."

"*If* he eats the meat of a tall animal," said the young man.

"That has nothing to do with it."

"My friend's father, who lives near the river, always eats the fat of the game they kill, and *he* is fat. So you see!"

"That has nothing to do with it," she said, and went into the cave.

The young man walked up and down angrily. "Why does she talk that way? Is she one of our sisters? I don't remember her."

"No," said the other. "She was with the people we fought with three seasons ago. She is my new mate and she is very good at magic, only I advise you

not to pay any attention to what she says."

The other picked up the scraping-stone she had left and looked at it with grudging envy. "The very tall man who killed the aurochs by himself has a son," he said, "and the son is short."

The other shrugged. "Don't pay any attention to her."

The woman came out again and looked at the sky, then went to the fire and stirred the pot with a stick. "I wish you would try to get the young animals," she said. "You always bring home the biggest ones and they are hard to chew unless I cook the meat all day."

"My father said that if you wish to be brave, powerful and swift, you must eat only the animals that are brave, powerful and swift," said the young man obstinately.

"Didn't he eat roots, too?"

"Yes."

"Well, then."

The young man threw the scraping-stone hard against the side of the cave opening and split it in two. "Roots are not animals!"

The young woman picked up the pieces and said, "I think I can make a small scraper out of the big piece and a throwing tip out of the little one, but this is a foolish way to get little stones. There are more little ones than big ones."

"Show him your bent stick with the animal sinew," said her mate. "She has a way of throwing very small spears with a bent stick," he said to his brother. He had a dim feeling that there should be peace between the other two, since they were near his cave, but he was scarcely aware of the feeling.

The young woman looked pleased and went into the cave and brought back a stick of springy wood with a thong attached to one end, and a few dried reeds.

"See," she said, and took a dried reed which had a small sharp stone stuck in the end of it. Then she bent the stick and strung the sinew from end to end. The younger man had his first view of a bow and arrow. "This was the way my mother showed me to throw little spears."

She fitted the arrow into the bow and, pulling it back, shot it at a pine tree on the other side of the fire. The bowstring twanged and the arrow wobbled, having no fletching, but it stuck into the tree trunk. The young man jumped back in alarm and blinked his eyes. Then he went to the tree and pulled at the arrow. It came loose, leaving the tip stuck in the bark.

"What good is it?" he said derisively, to conceal his astonishment. "It is a child's plaything!" He tried to pry out the arrowhead with his thumb, and broke his nail.

"If any child of mine played with this," said the young woman, "I should beat him." She put a larger arrow into the bow—one that had a heavier tip—and shot it into the same tree. Owing to its superior balance, this arrow did not wobble; it swished through the air and sank its tip deep into the soft wood.

"You have no child," said her mate, "so how can you beat it?"

The young woman said nothing, but she looked angry.

"My other women have children," he went on tauntingly. "They laugh at you."

"You have no child younger than ten seasons!" she said, and stamped her foot. "*That* is why I have no child! You are an old man!"

He started toward her with a look of furious intention. He had no spear in his hand, but he held a club with flint splinters stuck in the heavy head. She ran back to the cave mouth and put another arrow in her bow and aimed it at him. They both stood silently staring at one another. Then he threw down his club and turned away.

"Peace," he said.

"Peace," she replied, and dropped her bow. She went to the pot over the fire and sniffed it, poking at the meat with a sharp stick. "The food is ready," she said. "Will you take the pot off the fire? You have braver hands than I."

Q. How are we to find out anything about them, when you are so slow?

A. What are we supposed to find out?

Q. That is what you are supposed to find out.

A. I am to find out what I am to find out? You sound like them—like men.

Q. Like Man?

A. No, men. The women aren't quite the same. That's why I always choose to be one, but I wish you would send somebody else—another part of our Organism. I'm tired.

Q. Absurd. Besides, you are the best; you cannot be tired.

A. The best! How am I the best? You do nothing but criticize. You send me because I understand the intentions—the leanings—of live things. You say I understand understanding. I suppose that makes me some kind of epistemologist: the father confessor of the inscrutable.

Q. Wouldn't it be mother confessor?

A. Not with them; they don't like women to be priests. They can be holy, but they don't like women to tell them what to do. It's called nagging. They get especially angry if the woman is right.

Q. Hmm. Now you say that we criticize you. You surely are not going to claim to be above criticism *here*, are you?

A. Oh, no. I'm beneath it.

Q. Then why do you resent it?

A. Because it doesn't apply. If a mother is not a fool, she will correct her child, but she won't blame it. You can't go looking for good and evil motives in everything that happens. Does a stone have a motive when it falls to the ground?

Q. If this is the way you always talked, I'm not surprised you angered them.

A. I am sorry.

Q. You turn everything around that's said to you.

A. I will try not to. It's like the bishop—he complained about the same thing, and I was only trying to—

Q. What bishop?

A. I forgot his name. He was the thin one; he was much cleverer than the others. He gave me an impossible choice, so I chose to make another start.

Q. You mean you got yourself burned again?

A. Yes! They did it to all their best people. Both sides did. I would have looked a precious fool if I'd backed down.

Q. Can't you bear to admit you are wrong?

A. But I wasn't wrong. Anyway, they'd have made me out in the wrong either way.

Q. Did they only burn the women, when they thought the women were wrong?

A. No, of course not. And it was usually when they suspected the women were right. Then there were the women who were thought to be possessed by what they thought were evil spirits.

Q. They didn't suspect *they* were right, surely?

A. No, but they were afraid they might be. They were very unsure of themselves and their beliefs. That's when they burned people.

Q. It sounds very wasteful. They must be very careless of their possessions.

A. No, not in the least. I'll explain—

Q. I wish you wouldn't.

A. There, you see? They were just like you—they kept asking me questions and getting more and more enraged when I answered them. So, to shut me up, they tied me to a stake.

Q. You are too interested in your own reactions to things. Tell us about something more constructive—about what you found in other guises. I understand you led an insurrection?

A. If you call throwing an armed robber out of your house an insurrection. The trouble was that on that occasion a lot of my friends thought I was

right. That's called conspiracy....

The captain led a small group of foot-soldiers into the village at what, to the Romans, was the Twelfth Hour, which is sunset. The soldiers had light armor and carried only the small shields—not the enormous *testudines*—but they had been warned to keep their eyes open as the British were tricky, even treacherous. The captain greatly disliked to take such raw troops so far north, where the treaties were uncertain and the Pax Romana was held lightly, if at all. An ancestor of his, also a captain, had been killed near here in one of Hadrian's marches, and no one was quite sure what had happened.

The earth wall that Hadrian's men subsequently built across the British island was intended to keep out the more unruly natives of the North, and later the emperor Severus built another one of stone, but it was by now in a state of disrepair and only a few of the guard towers were manned. Even the Great North Road made an attenuated and unreliable line of supply, and the captain could expect neither reinforcements nor food from the camps further south, like Eboracum or Lancastrium.

Live off the land, he was told, and that meant quartering his troops—a risky thing because it separated them—or sending foraging parties to the surly farmers for "contributions." Since he was here to collect back-taxes, the inhabitants would not take kindly to feeding the collectors.

The village had a stockade of undressed logs and wide gates at either end. These were surmounted by arched wooden structures that were supposed to serve as watch-towers, but beyond spears and knives for hunting and the necessary farming implements, the villagers were not allowed to carry weapons of any kind. The stockade was not big enough to enclose all the houses, and the majority of these were on the outside and huddled against the walls.

The small body of Roman troops—barely a *manipulus*—were not surprised to notice that all the windows and booths had been shuttered, and in the exact center of the village, the local chieftain and heads of families were gathered in a respectful and anxious group. It annoyed the captain that it

was impossible to make an unexpected arrival anywhere in Britain; news traveled faster than Roman foot-soldiers.

"Hail Caesar!" said the captain, putting his arm up, the palm of his hand facing forward.

"Hail Caesar!" said the villagers.

"We come for the taxes which were not paid last year."

The villagers shook their heads and made regretful sounds.

"Nor the year before, nor the year before that. Which is your headman? I shall require food for my men at once—they are tired after a day's march."

A gray-bearded, very tall man stood forward. "The food will be ready at once, noble decemvir, and I hope you will honor me with your presence for dinner."

"Thank you very much," said the captain, "but I prefer to stay with my men until I see them taken care of. And I am not a decemvir. My rank is captain—Caesar's captain."

The bearded man bowed and said, "Then, after the arrangements have been made, Captain, will you not take a cup of wine?"

"Don't press him, Grandfather," a voice said from above them, and, looking up, the captain saw a girl's face at a second-story window. She had very dark skin, red hair and blue eyes. "If he's been walking all day, I expect he wants to go to bed early. You'll only keep him up all night talking about boar hunts."

"Silence!" the headman shouted. "Get back, girl! You insult our ... our guest!"

"No, let her stay," the captain said with an amused smile. "Better still, have her come down. I think I shall accept your offer about the wine later."

In the evening, the captain came to the headman's house with his two lieutenants as guard. They were received with deference and given wolf-

hides to sit on. The wine was brought by the granddaughter and served in horn cups.

"What is your name, young lady?" asked the captain politely. "This is excellent wine, by the way."

"Thank you, Captain," she said. "We have had a cask taken to your men. I made it myself, three years ago. My name is Boadicea."

"*Boadicea?*" said the captain in astonishment.

"No, no, Captain!" the headman said hurriedly. "She's joking—her name is Flavia; the other is the name she takes for herself. I apologize for her."

"It is not a joke," she said. "Boadicea is my heroine and I have taken her name. I don't like the name Flavia—it's Roman. Do I look like a Roman to you, Captain?"

"You look very beautiful," the captain said, laughing, "and there is no need for apology. I admire Boadicea myself; she very nearly drove Caesar's men into the sea. It was a long time ago." He drained his wine cup. "A long, long time ago."

"But we have not forgotten her, Captain," the girl said, filling his cup again.

"You insult our honored guest, girl!" her grandfather said. "Go to bed!"

"No, I beg you—please don't send her to bed," said the captain. "I'm not in the least insulted. After all, it's ancient history now. I don't think people think of us as conquerors any more. We are protectors. While we are here, the Picts stay where they belong, and the Scots, too."

"The Picts say they used to live hereabouts," said the girl.

"The Picts say, the Picts say! What do you know of what they say?" asked her grandfather.

"The cook's mother is a Pict," she replied.

"Well, she'd better not come *here!*" said the headman. "We want no Celts!"

"But, Grandfather, we *are* Celts!"

"No, girl, we are Romans," he answered, looking sideways at the captain.

The captain nodded. "That is true. All members of the Empire are Romans. Not citizens, perhaps, but Romans just the same, and all live by Caesar's law."

"But suppose people don't want to live by his law?" said the girl.

The two lieutenants looked shocked, but the captain smiled. "That would be most foolish and uncivilized of them. Don't you think it's better for the whole world to live as members of one community and cease all this useless warfare?"

"It seems to me," the girl said, "that warfare is the result of somebody trying to take somebody else's land and subject him to a law that is alien to him."

The captain raised his eyebrows and put his head to one side quizzically. The headman coughed and attempted to change the subject. "The taxes, Captain," he said, "are very much on my mind...."

"And on mine," the captain said. The two lieutenants tried to look businesslike, but they looked more as if they were falling asleep.

"And I hope I may say that this time we will have them ready for you," said the headman.

"I hope so, too," said the captain.

"But there are other levies that have not been made, which we had rather expected to be made...."

"Other levies?" The captain held out his cup and the girl poured more wine into it.

"I refer to troops, Captain," the headman said. "You levy no troops from us up here."

"You put me in rather an embarrassing position," the captain said. "You must realize that while I make no comparison to yourself, there are some people living at the outer boundaries of the Empire, people not yet wholly reconciled to Caesar's dominion, people who—to give another example—

think of themselves as, say, Helvetiae first and Romans second. It is the Imperial policy in such cases not to levy troops because—"

"In other words," the girl interrupted, "you think we are not to be trusted. It quite passes my understanding why anyone should expect loyalty unless it is freely offered."

"But, my dear young lady, you are not slaves! You are given the civilizing benefits of Roman rule, and you are taxed very much less than people living in Rome itself, I can assure you of that." He felt terribly sleepy—the wine was stronger than he had thought and he found it difficult to think of the right words. He was beginning to sound to himself like a senator, a race of men he secretly despised. "Let me put it this way," he went on. "A child does not *offer* loyalty to his parents—it comes by nature."

"Perhaps grown people do not like to be treated as children," she said. "I don't."

"You behave like one, Granddaughter!" the headman said. "Go to your room!"

Rather unexpectedly, she got up and walked to the door. "Good night, Captain," she said, but he did not answer. He was asleep and so were his lieutenants, and, since there were poppyheads in the wine, they did not wake up even when, an hour later, the shouting began outside.

Almost the entire detachment of the Roman troops was killed, and the captain and his lieutenants were being held hostage by the Pictish Decaledonae who had swarmed over the broken Wall—the break having been enlarged by the headman's granddaughter and her friends during the previous night.

The headman and his companions were horrified; they pleaded with the Pictish leaders to spare the Roman officers. "Caesar will send a legion," the headman said, "many legions to avenge this! Leave them unharmed and go back to the North, and the Roman captain will soften the blow that will fall on us all..."

The Picts told him to shut up and called for wine. The headman and his companions took advantage of the carousing to slip out the back way and, taking some of the villagers, including Flavia, they hid themselves in a cottage in the forest. Except for the girl, they were shaking with terror. She was triumphant.

"Now Caesar will withdraw again," she said. "He no longer moves north—but slowly southward. The next Imperial rampart will be below us, and we shall be free!"

"You are mad," her grandfather said. "Under Roman rule, we are safe. What can we expect from these Pictish barbarians?" He looked at her as though she were some new kind of snake.

"I should rather be occasionally robbed by my cousins than taxed to death by strangers," she said, her dark face flushed.

"But the Romans are civilized!" said her grandfather.

"Their civilization stands on slavery," she replied. "I'd rather be a free barbarian. The Romans are doomed."

"This is revolt!" the headman said. "In the name of freedom, you deliver us into the hands of the Picts—you are a traitor to your own people!"

"The Picts won't stay," she said. "They never do; they hate farming. What does it matter if they burn the village and steal some of the farm animals? It will come to less than what you would have to pay in taxes to Caesar."

"Caesar's men will return," said her grandfather, "and we shall have to pay ten times over. And if the Picts kill the captain, the Romans will have my life for it! You are a traitor! Who was with you in this infernal plot?"

Q. Why didn't you tell them? Why are you always so stubborn? You might have stayed on and found out many useful things.

A. There would have been nothing useful to find out. Men who submit to autocracy cease to be a living, growing organism. Look at Egypt—it stayed

that way almost uninterruptedly for four thousand years. However, I did find out one very surprising thing.

Q. I'm glad to hear it. What was that?

A. My grandfather was a Druid! I thought all that was dead and gone with the Roman occupation—but there was a secret sect and he was their high priest! So all the time he was in a conspiracy, too! I couldn't help laughing.

Q. How did you learn this?

A. They took me to an oak tree, put a wreath of mistletoe on my head, and he executed me with a stone sickle. Also all my friends who didn't have the sense to escape north over the Wall of Severus. But it made no difference in the end. The next emperor withdrew the army to the southeast part of the island and the next—or the one after; I forget which—took them all back to Rome. This was after we invited the Saxons in—they made it hot for Caesar's men, I can tell you! They also made things rather hot for us, but everything calmed down in time.

Q. It doesn't sound like much of an improvement.

A. Well, the Saxons may have been pretty bloodthirsty, but they hated slavery. They had sort of half-slaves—house-karls—but their heart wasn't in it. Also, although they were extremely rough, they didn't go in for official torture.

Q. But surely the civilized Romans didn't either?

A. I think you are being quite funny.

Q. I don't know what you mean.

A. I know you don't. That's the one really appealing thing about men: they sometimes have a sense of humor—when the joke is not on them. I think I must have caught it from them.

Q. Keep in mind that you are not an irreplaceable part of this organism!

A. How can I forget it?

Q. One gets the impression that Man felt that you were not irreplaceable either. When they want war, you are against it, and when they want peace—

like your North Britons—you are all for war. How did you hear about Caesar withdrawing from Britain?

A. I was supposed to go back a little later, but I missed again, and that time I was in real trouble—with both sides at once. It was just about a thousand years later, when the French and English were fighting each other.

Q. You seem to have made a rather dismaying number of mistakes.

A. I would never have learned anything if I had been afraid of making mistakes. Anyway, the bishops were the ones I had to fear the most, and when they started questioning me, I—

Q. Was it they who told you about the Saxons being invited to come in?

A. No, indeed. By that time, scarcely anyone knew anything any more, except prayers, recipes and how to supposedly cure warts. Later on, there was a revival and everyone became very clever, but I was in Italy at the time and I never got to hear about the Saxons until long afterward—my last trip but two, in fact. I was at a school in England....

The headmistress of St Agatha's prided herself on being fair. Her way of being fair was to avoid favoritism by being equally unfair to all the girls and to those of the assistant teachers who would stand for it. Some of them didn't, and they usually left after their first term, as the headmistress didn't believe in contracts. Besides, at the beginning of the twentieth century, contracts for teachers were a novelty.

The result of this policy was a rapid turnover in the young and intelligent teachers, and a small permanent staff of compliant sheep. That St. Agatha's had any scholastic standing was due to the fact that Miss Wakefield had taken honors at Girton, and the school's social standing was due to her being the cousin of a Peer of the Realm. The girls were fed almost enough, the school uniform was expensive, and nobody had much free time. French was well taught—by Miss Wakefield herself—and so was Latin, but games were also stressed. The school was run on what Miss Wakefield called the Honor System, which had the effect of dividing the pupils into tale-bearers and secret rebels.

On a raw November afternoon, Miss Wakefield sent a prefect for Sarah Stone, who was one of the new girls. "Tell her to come straight to my office. She can have her shower later," she said, and Sarah arrived in the jersey and serge skirt she had been wearing on the hockey field. Her bare knees were blue and her nose was running. She stood waiting while the headmistress looked with prominent eyes at some papers on her desk. Sarah could see that they were examination papers and one of them was in her own handwriting.

Without looking up, Miss Wakefield said, "I hear that your mother is in trouble with the police."

"But she—"

"Do not interrupt. I asked you no question and no answer is called for. It is a fact, which I have just read in the *Morning Post*, that your mother is in trouble with the police. Again—is that not true?"

"No."

The headmistress looked up in amazement "Do you mean to stand there and tell me the newspaper is *lying*? Do you tell me to my face that your mother is not involved with the ... the authorities?" Miss Wakefield also taught English Composition and woe betide the girl who used the same word twice in the same context. "We are blessed with the richest of all languages," she would say, "so let us explore it—let us make use of it—for to do otherwise would be tautology." She never made clear what tautology meant, but the girls got her drift.

"I don't know whether the newspapers are lying or merely mistaken, Miss Wakefield," Sarah said, "although my mother says that it's hard to tell the difference with most journalists. At any rate, she is not in trouble with the police. They are the ones that are in trouble."

The headmistress stared hard at Sarah; she was rather good at this with small girls of thirteen. (You and I might find it difficult to stare down a child, and impossible in the case of a kitten, but Miss Wakefield was, after all is said, the cousin of a Peer of the Realm.)

"I believe I can understand that," she said. "In fact, I pity the arresting officer. Here is a woman who breaks shop windows for the sake of attracting attention to her political *clique*, and he is no doubt subjected to

scratching and biting. Votes for women, indeed! Does breaking shop windows prove that people like her should have the ... the franchise?"

"She didn't break the window," Sarah said. "She was pushed against it by the policeman. And she never scratches unless a mosquito happens to—"

"You were not there, Stone," said the headmistress, "so how can you say that?"

"I know my mother. And she doesn't bite, either," Sarah said, looking at Miss Wakefield's neck. "Unless it's a tough old hen!"

Miss Wakefield had enough sense to refuse the bait, but she flushed. "I do not feel that it is at all suitable for the mother of one of our girls to be a Militant Suffragette! The reputation of the School...." The sentence was left unfinished.

She picked up the sheets of paper. "I have here two of the mid-term examination papers in arithmetic, yours and Angela Harvey's. There is a curious, a very curious similarity between them. All the answers are correct except for problems five, seven and twelve, and they have precisely the same mistakes in both papers!" She paused and stared hard at Sarah, who blinked but refused to lower her eyes. "You and Harvey sit next to one another," Miss Wakefield said meaningly.

Sarah said nothing. She sniffed because her nose was running and there was no pocket in her games uniform for a handkerchief.

"Well?" said Miss Wakefield. "Have you nothing to say?"

"No, Miss Wakefield," Sarah said, "except I didn't copy from Angela, if that's what you mean."

"Then it would appear that she copied from you."

"That's a beastly thing to say! It was a coincidence! She's not a cheat!"

The headmistress felt on secure ground: the child was losing her temper. It was Miss Wakefield's favorite stratagem to make people lose their tempers—that is, if they were children or underlings.

"Blow your nose, Stone," she said, and then, seeing that Sarah had no handkerchief, she gave her her own, with a look of distaste. "I think perhaps you might do better at some other school."

"So do I, Miss Wakefield," Sarah said. "Mother wanted to get me into Mr. Russell's school, but it was full up."

"*Bertrand* Russell?"

Sarah nodded, blowing her nose again. She was shivering.

"*Well!*" said Miss Wakefield. "I never heard the like! He's an Atheist! Why, he believes in *Free Love!*"

"I don't know what he believes," Sarah said. "I know he was awfully nice when he came to tea. He said I had some kind of a guiding somebody standing over me. He said he would like awfully to have me at his school, but it was full up. I know one of the boys there and he says it's simply ripping."

"*Well!* Of course, if your mother thinks of us as Second Best.... Perhaps Mr. Russell believes it is all right to cheat in examinations, but we have a Tradition at St. Agatha's." She rang a bell on her desk and a scrawny little housemaid came in. "Send one of the girls for Angela Harvey," Miss Wakefield said. "Tell her to come here directly." The little housemaid bobbed respectfully and went out. "Now we shall see what *she* has to say," the headmistress said.

"She'll only be frightened and cry," Sarah said, "and she'll say anything you want her to. She wouldn't *dare* cheat in an examination."

"Then you admit that you copied from her?"

"I do not!" Sarah said, her teeth chattering. "I tell you it was a fluke! Miss Somerville jolly well knows I wouldn't do it!" Miss Somerville was the new and still enthusiastic math teacher, but her enthusiasm would be gone by the end of the term, and so would Miss Somerville.

"That will do!" said the headmistress. "Impertinence will not improve matters."

There was a timid knock on the door and a girl of Sarah's age, but smaller, came in. She had changed into the school uniform and wore steel-rimmed spectacles.

"Stand beside Stone, Harvey," the headmistress said. "Now I want you to think very carefully before you answer what I'm going to ask you."

Angela Harvey looked terrified and began to cry.

"There, you see?" Sarah said. "You're only doing this because you don't like my mother! You want me to leave school, and it's the only excuse you can find!"

"Be quiet," Miss Wakefield said with an unpleasant smile. She never lost her temper. "Did you, or did you not," she went on to the damp Angela, "copy the answers in your arithmetic from Stone?"

"Oh, no! Oh, I *wouldn't*, Miss Wakefield!"

"Then how is it you have seventeen right answers? You never do as well as that, and you got the same three wrong that Stone did."

"I don't know, Miss Wakefield! I don't *know*!" Angela sobbed loudly and became smaller than ever.

"I'm afraid," said Miss Wakefield, who looked quite otherwise, "that unless your *friend* here can explain this curious—this odd coincidence by admitting she copied *your* answers, I shall have to ask your parents to remove you from St. Agatha's at once."

Sarah's face was bright red, but it had the look of fever. "How simply rotten of you! You're just trying to get me to confess to something I didn't do, to save Angela!"

The headmistress felt her heart beat with excitement and pleasure. Why, the child was positively crimson with temper! "You are not helping her by behaving like a common guttersnipe. At this school, we try to behave like ladies. Perhaps at Mr. Russell's—"

"At Mr. Russell's school," Sarah interrupted, "I'm sure nobody would think it was worthwhile to cheat."

"Then you admit you cheated?"

Sarah looked at Angela, and back at the headmistress. "Yes!"

Miss Wakefield smiled. "Well, then, I think there is nothing more to be said. You may go, Harvey."

"You," Sarah said, looking at Miss Wakefield with blazing fury, "are a coward and a—a *black-mailer*!"

Tiny cracks seemed to appear in the headmistress's porcelain composure. Angela had not yet left the room and heard Sarah's outburst. She stopped at the door and turned around with wide eyes.

"Go at once!" cried the headmistress to her, and waited until the door closed. "You are to be expelled publicly from the school!" she said to Sarah in a low, unsteady voice. "And first you will be publicly thrashed!"

Sarah's face was patchy now, red on white, and her skin looked dry as paper. "If you touch me, I will kill you. I'm not afraid of anyone like you. I didn't cheat in the exam. I said it to keep you from expelling Angela, and you knew it all the time. Everything you say is a lie. You just want to get rid of me because of my mother. You are against votes for women because you are a liar. You told us in history class about government by consent, but how can it be when half the population have nothing to say in the matter? I'm going to pack and leave, and if you try and stop me, I'll...."

She went fiery red, and then white, and fainted.

The headmistress was breathing hard, and later, when Sarah was taken to the san, she was frightened. Sarah's temperature was 107 and she had the most virulent kind of pneumonia the school doctor had yet come across. He was almost more curious in watching the course of the disease than he was concerned with the patient, but he did not have very long to watch it, for Sarah died shortly before sunrise.

Q As far as one can follow your line of reasoning, you claim that the head woman of your school was untruthful, but was against untruth.

A. Yes. Quite a lot of them are.

Q. They sound mad.

A. Well, they are and they aren't. They lie to themselves, mainly; that's what causes most of the trouble. They have a saying: Know Thyself, but nobody ever—

Q. *They* have? Who said it?

A. All sorts of people are said to be the ones who said it first, but actually I think I was. I was living on an island in the Aegean Sea, and the mainland Greeks thought women shouldn't be writing poetry, so there was a row about it. They said I invented hexameters—which was nonsense—and that made them angry for some reason. So, later on, they decided I was a myth.



Q. Is this the Sappho you mentioned earlier in this hearing?

A. No, no. She was later and she didn't become a myth. My name was Phemonoë. I meant to tell you about that trip. My father was—

Q. Never mind. We've heard enough of the early trips. What we should like to hear about is your last. A decision must be made about these people—we've waited long enough. While it must be admitted that you are the best we have for the task, you not only take a long time and make error after error, but in the very process of examining them, you alter the subject of examination.

A. Yes, I know. They have a new phrase for that. They call it the Uncertainty Principle. For example, you can't determine the mass and velocity of a particle and at the same time its position. If you measure the one, you alter the other.

Q. We are quite aware of that.

A. I just thought I'd remind you.

Q. Unnecessary.

A. That's what men usually say; they dislike being reminded. Am I to stop making trips?

Q. That will be decided in the light of the rest of your report. I may tell you now that there will probably be no further trips. You will be reabsorbed into the Unity.

A. I see. I remember you said the same thing after I reported on the time they hanged Haman. You seemed to side with him. Anyway, if I get reabsorbed, it won't be a Unity any more—not the way things are going.

Q. You overrate yourself. Contact with Mankind has changed you.

A. Oh, it has! I've changed them a bit, but it's the principle of uncertainty again: it changes me, too.

Q. The Unity is greater than its parts.

A. Not if it's infinite, the way you say it is. You know, it's a funny thing, but I've never been quite clear just what's behind all this decision you talk about. What is our purpose?

Q. Does a stone have a purpose when it falls?

A. I'm not talking about values. What are the alternatives you imply in the decision?

Q. There are three. We destroy them; we absorb them; we ignore them.

A. I'm afraid they can't be ignored.

Q. Why not?

A. It's too late. The Unity should have started ignoring them right at the beginning—we are already changed. And if they are absorbed, we shall be still more changed.

Q. *They* will be changed. The Unity is eternal and—

A. You ought to talk to a man called Heisenberg. He called it the inexactitude principle, but it's the same thing. For example, men are always going around asking each other questions; they call it taking a poll, only when you try to find out that way what people are thinking, you change them. Or anthropology—when you study a tribe, you alter its way of life. Furthermore, it alters yours.

Q. It would appear that you have lost your sense of objectivity.

A. That's the way my last husband talks. There is no such thing. It's a strange fact, but it seems that the mathematicians are the only ones who have a glimmering of the truth—they and the physicists. I was beginning to think that mankind as a whole was progressing quite nicely.

Q. I thought you said they were. It seems you're never satisfied.

A. Well, some things improve, but their point of view keeps changing with regard to what should and what should not improve. It's hard to say whether the Greeks really believed in progress: they thought there had been a golden age and that the world had degenerated from it. Some of them may have wanted to return to it, but I always suspected their motives—by their own showing, they were decadent. During the Middle Ages, it was felt that art was on the way up—part of an evolutionary process—whereas science was

not. Aristotle and the Thomists had science all cut and dried. Nowadays it's fashionable to say the art was as "good" in primitive times as it is now, while science on the other hand is evolving to a higher state of truth. The latter happens to be true, but they still have war.

Q. Perhaps it's inevitable.

A. If it is, we are wasting our time.

Q. That is for the Unity to decide. You set yourself up as Mankind's conscience.

A. Not conscience. I plead for self-examination—for a reappraisal of ideas.

Q. Yet you only succeed in irritating them.

A. That may be the best way. And you confuse conscience with consciousness. If there's one thing I've found out, it's that Man differs from the animals in having more consciousness, just as animals have more than plants. I don't suppose that hydrogen has any at all.

Q. But you have turned what was intended to be a field-trip for examination and analysis into a crusade. With all your nagging and irritating them, there have been no results—no real advances.

A. I thought you were complaining that I was altering what I was sent to examine. You talk about unification—or absorption—as if it were a catchword. That's the trouble with generalities: they're not necessarily true in all cases.

Q. You mean they are too general?

A. I mean that they are not general enough. I agree that men progress too slowly toward unification, but we mustn't confuse it with domination. We cannot *impose* it on them. That would lead to a world divided into the ruled and the rulers—not a unity.

Q. Then you are for absorption?

A. You know, you twist things around much worse than I do.

Q. The Unity is incapable of—

A. Furthermore, I think you have been altered more than I have.

Q. You are part of the Unity.

A. And the least altered part. You won't be able to absorb them the way you can reabsorb me without destroying them as entities.

Q. You set yourself up as the only one to know this. Why?

A. Because I have been the one to make the trips. I have been your eye.

Q. But the others—the ones you called the spies?

A. They weren't there to look at Man, only to watch *me*. They weren't even sightseeing—they were slumming. However, I think I am ceasing to be the only one. I think you are coming to know these things, too.

Q. Very gratifying. Now, as to the latest trip?

A. There seems to have been a slip-up....

Q. *Another* one?

A. Different. The ones *I* made were errors in time; this one is not mine, and it's in hyper-time. I was trying to explain it to a friend, but he already knew all about it and that led to the slip-up. It caused it, yet it came afterward.

Q. How annoying for you. How did you explain hyper-time?

A. I said that when an object moves or changes, time is needed as one of the coordinates to describe that change. I said that consciousness moves *through* time—from Monday to Tuesday—otherwise we would be merely aware of differences without experiencing them as change. I said that to describe this motion of consciousness along the dimension of time, *another* coordinate is needed: hyper-time.

Q. And the slip-up—which you claim is not yours?

A. Is in hyper-time. It is the result of the Unity and Mankind affecting one another. You have, through my efforts, examined them—and thus changed them. Now they begin to examine you—with the result that *you* change.

Q. They begin to examine *us*? You must mean they have examined *you*.

A. There is a man—a young physicist—and he has found out something. I think that without quite knowing it, he has detected you. At all events, he has found out where you are, and I think that perhaps you are aware.

Q. What makes you say that?

A. Obviously these things work both ways. Heisenberg's principle says—

Q. We want to hear no more of Heisenberg's principle! There's enough confusion as it is, without that!

A. I admit it. That's why I decided to—to close my eyes to everything but essentials on this trip.

Q. It is gratifying to hear you admit something for a change. What are you "closing your eyes to" in this case?

A. Appearances.

Q. Why?

A. Appearances are deceitful. That is, they are now; they weren't before, when the Unity was the Unity and Mankind was Mankind, not something of each. You ask me to keep my objectivity and you don't tell me how. You can't, of course—your own is too lost for you even to know it's gone. So I have to work out my way alone and the best method seems to be to work with as few senses as possible. That won't give me real objectivity, but it will mean somewhat less involvement.

Q. The less you see, the more you can observe? Does that make sense?

A. Nothing does any more. Oh, if you had only stopped in time—no, that wasn't possible.

Q. Why not?

A. Because, being in hyper-time, the slip-up is both in the past and the future in simple time. The last trip is going on now....

Katherine was lying on the lab sofa with her hands behind her head. The sofa was shabby and was alleged to have belonged at one time to a psychoanalyst. Its present function was to offer temporary rest to anyone working late in the lab. Today was Sunday and no undergraduates were there.

"What are you fiddling around with, Phil?" Katherine asked.

"The electron microscope," he said. Phil Kaufman was an assistant physics professor, short, bony and intense-looking, and at the moment he was engaged in extra-curricular research.

"You know, I bet this old chaise-longue could many a tale unfold," she said.

"Well, according to rumor, many have been unfolded on it."

"Professor, your mind wanders. I'm thinking of its previous condition of servitude. Think of the dreams it used to hear."

Phil Kaufman didn't answer. There was a pause and she said, "This afternoon you're working with the microscope, and last night it was the telescope. You were in the observatory until dawn."

"How did you find that out?"

"I have my methods, Watson. I don't see how you expect to keep going on no sleep at all. Russ is worried about you."

"Pro or con?"

"Pro, of course. He likes you very much. In fact, he thinks you are the best brain on the faculty."

"Coming from the president, that's praise indeed." Phil got up and went to a desk, where he looked at some notes. "Speaking as my boss's wife, would you say he was pro or con about this work I'm doing?"

"I would say he can't make it out. Alternating between the Microcosm and the Macrocosm. Incidentally, why don't they call that thing in the observatory a macroscope? I don't think Russ is very good at understanding the unfamiliar. I was telling him about the concept of hyper-time the other day, and his reaction was one of solicitude—he got me a drink." Katherine stretched her arms. "What are you doing now?"

"Checking some figures. You know, that was odd, your bringing up the business of hyper-time. This thing I'm working on seems to involve it."

"Oh?" Katherine put her arms behind her head again. "Tell me something, Phil—what does he look like?"

"Doctor Russell Farley?"

"Yes. I suppose it's a funny sort of question to ask about one's husband. How does he look to you?"

"Like the youngest college president in America, I guess. Brawny but brainy. You make what they call a handsome couple."

"Yes, I was going to ask you what I looked like, only it's a waste of time. People never tell you."

"I can," Phil said, "but I won't, for fear of giving you a swelled head."

"As a matter of fact, it's silly of me to ask," she went on. "I wouldn't understand. I don't even know what 'pretty' means, although I have a dim idea what 'ugly' does. Color is another enigma to me. Somebody once told me it's like a smell, but when I get a bad cold, I can't remember what smells are like. It's like not being able to think of the word 'bubble' when your mouth is wide open—you think of 'Ah-uh.'"

"I'll tell you one thing about yourself," Phil said. "You don't look as though...."

"As though I was blind?"

"Correct. And it's incredible the way you get around. You never bump into anything, and you look people right in the eye when they talk to you."

"They say it's hearing faint echoes from an obstacle—like a bat. Personally, I *feel* the wall in front of me. I admit when my ears are stopped up I can't hear the wall, but I'm not so sure that's a convincing proof. It's the same with pit vipers—some smart investigator discovered that when you plug up their little heat-detecting organs—I guess those are the pits—they can't locate warm prey in the dark. Conversely, in the dark and not plugged up, they will strike at a hot-water bottle."

"Sounds pretty convincing to me," Phil said, and went back to the electron microscope.

"Tush," Katherine said. "How about people not wanting to smoke in the dark? Does that prove that the sense of taste depends on sight? *I* smoke. In

fact, you might bring me a cigarette and an ashtray. The only reason most blind people don't smoke is they're afraid of fire." She took the cigarette Phil brought her. "Thanks."

"Aren't you afraid of fire?" he asked.

"Of course not. I can detect a match flame at fifteen feet."

"You ought to go to Duke University sometime and have Rhine take a look at you."

"I did. All he said was 'Hmm,' and I joined the other statistics."

There was silence for a while, interrupted at one point by a muffled "Damn!" from Phil peering into the electron microscope, and the warm sun lay across Katherine's lap. Finally he straightened up and switched off the current. "Well, it's there, all right," he said, and got up and went to the couch and sat at her feet.

"What is?"

"The red shift."

"Aren't you confusing things?" she said. "You're not in the observatory now, Buster; this is the lab. I thought the red shift was the recession of the distant galaxies ... whatever 'red' is."

"Quite right, Holmes. However, in this case, it's the recession of the not-so-distant atoms. They are small-sized solar systems, too, in a way, and when I say 'red' I mean something I can only infer mathematically, because I'm not dealing with light in the ordinary sense."

"You mean they're *receding*?"

"Only in this context," he said. "Motion is length over time; in this case, it's length over hyper-time, so they're still here in the lab."

"I'm relieved to hear it," she said. "However, I should think they'd be receding into tomorrow."

"They are, but into yesterday, and new ones from tomorrow are continually coming in to take their place. It's like Fred Hoyle's theory of the continuous birth of hydrogen."

"You're making me feel like my poor husband," Katherine said. "I understand the necessity of hyper-time to describe the motion of consciousness along time, but what's this got to do with the atoms?"

There was a knock at the door and Phil stood up, just as Doctor Russell Chalmers Farley came in without waiting for an answer. Phil and Katherine felt faintly embarrassed—there was scarcely any need to knock on the door to the physics lab; it somehow suggested that the door should be kept open when entertaining callers.

Doctor Farley was a handsome man of thirty-eight with a blond mustache that gave him the look of a Kipling colonial officer.

"Ah, there you are, Katherine," he said cheerfully. "Hello, Kaufman. How's the Research Magnificent?"

"It's beginning to show signs of life," Phil said. "I think I can detect a sort of fetal pulse."

Doctor Farley blinked his pale eyelashes and smiled. He sat down at the end of the couch where Phil had been sitting and looked up at him. Part of his charm was that, when he talked to a man shorter than himself, he got below him and looked up. At his evening "sherries" at home, he had a way of deferring to the newest and least important visitor, who was thus raised to the temporary rank of philosopher, while Russell Chalmers Farley was reduced to the position of listener.

The role of humble servitor of the Truth was his most useful one—it had worked rather well with Katherine, and he had an adroit and imaginative way of expressing his ideas, which usually disguised the fact that they were generally borrowed.

They had met at a street corner in New York City where she was waiting for the sound of traffic to abate so that she could cross. He was on the opposite side, and with his extraordinary eyesight and intuition instantly recognized that the beautiful, odd-looking girl facing him on the other side of the street was blind. He was at her side before the light—and the sound of traffic—

had changed, and said, "I hope you don't think I'm being forward, but let me offer you my arm. Taxis have a way of making illegal turns sometimes...."

"You are very kind," she replied, pulling him back from the path of a taxi making an illegal turn. "You have a very nice voice," she said as they got to the other side. "I guess being blind makes one ... forward!" She laughed and started to walk on.

"No, please wait!" he said, and caught up with her. "I wish you hadn't said that. It can be taken in another way: that I am forward because you are blind. I should like to say that *you* have a very nice voice."

She stopped and laughed again. "That's one of the nicest things I've ever had said to me!"

"Do let's ... I mean would you let me...." He floundered, and laughed, too. "Can't we have a drink together? Now?"

"I think it would be lovely," she said.

Later on, he said to her, "You may think this impertinent of me, but you make me envy you. If I were braver, I should wish that I were blind. You actually see more than I do."

Katherine was intrigued. She had been told this before, but always with mystical and pseudo-religious implications. This man, with the attractive voice and smell, had no trace of the mystic.

"Let me tell you a fable to illustrate what I mean," he went on. "There was a man who was born blind, and he went to work as a coal miner because the darkness was no hindrance to him. One day while he was working alone in an unlighted gallery, his sight was miraculously given to him.... He shouted out in amazement and awe, and the other miners came stumbling to him in the total darkness.

"'What is it?' they cried. 'What's the matter?'

"'I can see!' he told them. But they were puzzled, for they had brought no lights.

"'What can you see?' they asked. 'There is nothing to be seen here in the dark.'

"'I see *black!*' he said. 'In front of my face is blackness—however, at the back of my head, I'm still blind and I see nothing.'"

Katherine was delighted. "I'm not quite sure I understand."

"Why, to the blind there are no shadows," Farley said. "Another drink?"

"I think it would be lovely," she had said, and since she could see no shadows, she had begun to fall in love.

Doctor Russell Chalmers Farley looked up at Phil and smiled. It was a charming smile and it was as genuine as a guaranteed, ten-carat, real, honest-to-goodness zircon. "As Katherine has probably told you," he said, "what you are doing is completely over my bowed head. I am enormously impressed and at the same time unable to comprehend."

"I find it hard to comprehend, too," Phil Kaufman said. "And I suppose that's what leads me on."

"Well, the thing is," Farley continued, "Washington seems to have gotten wind of it, and you know how they are. They don't like things to be over their heads."

Phil Kaufman looked at him in astonishment and sat on a lab stool. "I don't understand. How can they possibly be interested in what I'm doing? It's purely theoretical research."

"Surely you don't deny that Lisa Meitner's researches began by being theoretical? And look what *they* led to. The point is, Kaufman, that I have been informed that we are about to receive a visit from a man from the A.E.C. He's arriving here sometime this afternoon."

"But that's absurd! I'm not *doing* anything to atoms. I'm merely examining them!"

Katherine frowned when he said this. Phil knew better. Worse yet, so did she.

"When the A.E.C. hears of somebody working in atomic research," Farley said, "they want to know what's cooking. I hate you to be subjected to this,

but it won't do any harm to be polite to the fellow and let him, as it were, look over your shoulder."

"I'm damned if I see why I should!" Phil said. "What does he expect to do? Classify me?"

Farley laughed placatingly. "I know it seems high-handed, but I think we all ought to remember there is such a thing as Security."

"Security, my foot!" Phil said. "It was that kind of demented thinking that caused Germany to lose Lisa Meitner! *And* Einstein."

"What strikes *me* as rather odd," Katherine said, "is their sending someone here on a Sunday. When did you hear about it, Russ?"

"A little while ago. On the phone."

"Curiouser and curiouser."

"He was very polite and apologetic."

"Quite typical," Phil said. "It's the velvet-glove touch."

Farley looked at his wristwatch. "He won't be here for a while, so I wish you could brief me about the inwardness of what you are doing, Phil." He'd never used his first name before, and Phil became a little wary. "I know you can't give me a ten-year course in advanced physics this afternoon, but—well, I'd like to know what kind of stand to take. I'll be representing the university, after all."

Phil Kaufman looked down from his perch on the stool at the earnest, kindly face and wondered what really lay behind it. So far as he could see, Doctor Farley had no reason to take any stand on the question at all, except to tell the A.E.C. man to go sit on a tack. If he wanted to represent the university, let him do it in the name of Academic Freedom. Phil glanced at Katherine. She was sitting very still and he had the impression that she was thinking about something else.

"All right, I'll give it a try," he said. "There's an idea that's been around for quite a while that there is an analogy between the stars and the atoms."

Doctor Farley's face lighted. "I believe I've heard of it. Back in the 'twenties, by a man called Dunn, wasn't it?"

Phil shook his head. "Twenty years earlier by a man called Fournier-d'Albe. He wrote a book called *Two New Worlds*, in which he suggested that the solar systems are actually atoms in some vast cloud of super-gas. Of course, this notion ignores the celestial absence of molecular structure—unless you count double stars as molecules—but it might be accounted for by assuming a high temperature. Then he said that the newly hypothesized Rutherford model of the atom was a sub-microscopic solar system, but he didn't stop there.

"The atoms and their electrons, he said, were in turn made up of sub-atoms and were perhaps populated by sentient beings who looked through their telescopes and counted the atoms in their vicinity, no doubt arranging them into constellations. You can carry this imaginary process in both directions and as far as you like, but are we to decide arbitrarily that it goes on infinitely? Or is it like Einsteinian space, finite but unbounded?"

"I have asked myself this question and I believe the latter statement to be in a sense correct, but what does it mean? Well, it means that if you move further and further into larger universes, you eventually get to where you started. Not that Big is the same thing as Small, but that from wherever you happen to be, the ones in the direction—outward—look successively bigger, while the ones in the other direction—inward—look successively smaller. Now if there were some kind of super-telescope that could look beyond our universe of super-atoms, and beyond the next and so on indefinitely, you would find yourself staring up through a super-microscope at your own eye."

"Get along with you!" Katherine said. "This is the pipe dream to end all pipe dreams. Tell us more."

"Well, I'll revise it to this extent," said Phil. "It wouldn't be your eye that you'd see, any more than you'd see your own face if you looked far enough across ordinary intergalactic space. You'd see the back of your head—or, rather, the other side of the Earth—provided there was nothing in the way."

"And in this case it would be what?" she asked.

"I don't know," Phil said, looking worried. "What is the equivalent of the back of your head—looked at along the direction of hyper-time? Could it be that what you saw would not be from behind, but from ... inside?"



Katherine's beautiful sightless eyes seemed to be turned inward, and she sat very still. Then she said, "You evoke something in my mind like the echo of a picture I once knew, and will know again."

Farley looked at her sharply.

"You mean something in your subconscious?" Phil asked.

"Perhaps that's what it is, and yet they say that you try to escape knowledge of your subconscious—that it frightens you. I am not frightened, Phil. I feel ... expectant."

"I'd feel more expectant," he said, "if I were quite sure of what I was doing. The trouble is that while ordinary light could in theory show you the super-astronomy of the stars and planets that are made up of atoms consisting of our stars and planets, it won't work the other way."

"Why not?" Farley wanted to know.

"Wave length. As it is, we have to use an electron microscope to see the larger molecules; the wave length of visible light is too coarse-grained to show anything that small. So just try to imagine how impossible it would be to see the sub-atoms—infra-atoms—that I'm talking about if one had to rely on ordinary light! The electron microscope wouldn't help, either. It would be exactly as though some gigantic, super-researcher were trying to look at one of our molecules by bombarding it with a shower of planets."

"Then how can you see this 'red shift'?" Katherine asked.

"I can't," he said. "I detect it by a kind of mathematical diagnosis. It's an inferential process—as most forms of observation are, in modern physics."

Farley was looking as intelligent as he possibly could, but it was plain that he was out of his depth. He had heard of the red shift, but he decided he had better not have it explained.

"There's another thing," Phil said. "The time it would take light to make the round trip of our Einsteinian finite universe would be so great—in the order of $4\pi \times 10^8$ years—that not only would you not see your not-yet born self, but the Earth wouldn't have been formed either. The light you saw would be that many years out of date. However, in this case the elapsed time would be hyper-time, and you'd be there in ordinary time."

Doctor Farley got up and walked to one of the windows and stood looking out at the observatory across the campus. "Am I to understand then," he said, "that you are trying to formulate a new atomic theory?"

"Not in the sense of in any way modifying the accepted one," Phil said. "If I'm right, it will merely be a new way of looking at the Universe as a whole, and it won't have the slightest effect on anything."

"I should have thought," Katherine said, "that being able to see inside one's own head would have all sorts of interesting effects." She got up. "I've got to get back to the house. We've got people coming to dinner, Russ, and I'd better get things organized. Are you coming?"

"I'll be along in a little while, Katherine," he said. "I want to hear more of what Kaufman has to say." He refrained from guiding his wife to the door because of long habit, and again sat down on the couch. After the door closed, he and Phil listened to her sure footsteps going down the corridor. They looked at one another a little guardedly.

"You know I'm on your side," Farley said when they could no longer hear Katherine's footsteps. "Surely you know I don't like this any more than you do, Phil."

"I suppose you don't."

"You won't mind very much if I ask you a favor, will you?" Farley said. Having asked a rhetorical question, he seemed to be illogically waiting for an answer. Phil was unaware of the chess game, but wondered uneasily what was coming.

"Will you please leave her alone?" Farley said.

"I—" Phil started to say, but Farley held his hand up, palm forward.

"My dear chap, you are one of the most sensitive and kind people I know. But you are a little thoughtless. You imagine that, because Katherine is blind, you are doing her a favor by—by giving her companionship. You feel that her interest in the world can be furthered by your interest in her. This is not the case. I ask you please to leave us alone."

"Us?"

"Yes. You put me in the embarrassing position of having to say that we are very well as we are. I know that Katherine is impressed by your—your mind, and I know that your sympathy is well-intended, but it is misplaced. She needs no sympathy."

"Why not?"

Doctor Farley spread his hands, a gesture usually meant to substitute for words. "Do the strong need sympathy?"

"I think so," Phil said.

Doctor Farley smiled. "Well, then, think of *me* as the strong one—the one who needs sympathy as the guardian of something precious. Will you give me your sympathy?" He smiled still.

Phil realized that when the A.E.C. man came—when any pretext presented itself—Doctor Farley would throw him to the wolves.

"Katherine is not in love with me," Phil said.

"But are you with her?"

"No. At least ... I don't know."

"Then you are."

"Aren't you being a bit old-fashioned?" Phil said.

Farley had abandoned his usual pose of sitting and looking up. He looked down at Phil—in fact, he looked down his nose and past his blond mustache.

"I mean," Phil went on, "I think Katherine ought to be the one to decide whether she wants to go on seeing me."

"And I disagree."

"And I," Phil said, "shall stop seeing her when—and only when—*she* wants me to! I refuse to be ordered around like this. We're not doing anything wrong!"

"I think you're forgetting—"

"I'm forgetting nothing!" Phil interrupted. "You're acting as though I were having an affair with your wife, and you're trying to pull rank on me! I don't intend to be browbeaten and threatened!"

"I'm not threatening you, my dear man," Farley said, his eyebrows raised. "I ask you as a favor, that's all. I think I know my wife's—mind better than perhaps she does herself. And certainly better than an outsider can."

"And you regard me as an outsider?" Phil's voice was loud.

"You know perfectly well what I mean!" Farley replied angrily. "You are not her husband and consequently do not know—"

"I know her a damn sight better than you do, you stuffed shirt!"

Like most blond men, Farley became red very easily. At the moment, he resembled a tomato with yellow hair. "Why, you little—"

"*Really!*" At the sound of Katherine's voice, they both swung around. They had been making too much noise to hear her return, and she stood at the open door. "Isn't this a bit undignified?" she said. "I could hear you outside."

Farley was breathing heavily. "What brought you back, Katherine?" he asked, finally.

She walked past them to the psychiatric sofa and sat on it without answering the question. She looked as though her mind was on something else—and then, suddenly, startled and intent.

Yes! I am here....

Neither Phil Kaufman nor Russell Farley heard her—they were intent on avoiding one another's eyes, but they would not have heard her anyway.

Q. You were right. He knows where the Unity is—if not what it is, yet.

A. Oh, he will.

Q. Are you so sure? And will you at last admit that we are right?
Unification—it's the only way ... now.

A. (she has her face toward the electron microscope; her blind eyes seem to probe it) One cannot impose it on them. What kind of unity can come from imposition?

Q. And are things to go on as they are?

A. No, it's too late. Things have already changed....

Q. The history of Man has been the history of his integration—from families to tribes, to communities, to city states, to nations, to hemispheres, to—what next? Is it to stop here, and the hemispheres to beat each other down to the tribal or family level?

A. You will be destroyed in the process.

Q. *We*? In the process of unification?

A. Of course.

Q. And you?

A. I'm always being destroyed.

Q. Ridiculous. Unification can scarcely destroy the Unity.

A. If you unite with Disunity?

Q. The decision has been made: Absorption.

A. By whom? Of whom?

Q. The joining of the collective subconscious to the mutually antagonistic egos of all men. Freud of Vienna had this as a goal—you told us that yourself and I quote it back to you.

A. Or the reverse—men's mutually antagonistic egos in combat with the Unity?

Q. We will take that chance. Now watch—look at the world around you and you will see a dominion of universal brotherhood, the moment Unification is imposed!

A. I will look, but is that what I'll see?

Q. Now! Look!

She looked at Phil and then at her husband, who looked back at her questioningly.

"You were going to say something?" he asked.

She shook her head, and he shrugged his shoulders.

"This business I'm working on—" Phil began, and hesitated.

"Oh, yes. That reminds me," Farley said. "How's the Research Magnificent?"

"It's beginning to show signs of life," Phil said. "I think I can detect a sort of fetal pulse."

Dr. Farley blinked his pale eyelashes and nodded. He sat down at the end of the couch where Phil had been sitting and looked up at him. "Well," he said, "I just thought I'd drop by and see how you were doing. I'll never be able to understand it, though."

"I was going to say, do you think the A.E.C. might conceivably be interested?" Phil said. "After all, it is sort of vaguely connected with atomic stuff."

"I can't imagine why they would be," Farley said, and glanced at Katherine. She had gotten up and was standing at the window.

"The sun's going in," she said, "and it looks as though it may rain. I've got to get back to the house." She turned around with a smile. "How about having dinner with us tonight, Phil? We've got some people coming who'd like to meet you. Don't you think that would be nice, Russ?"

Dr. Farley didn't look as though he thought it would be nice at all, but he said nothing, and neither did Phil Kaufman.

"If you're coming, you'd better straighten your tie," Katherine said. "It's under your ear, as usual."

Phil reached up absently and pulled at it with one hand. "Sorry."

"You put me in an embarrassing position," Farley said. "I think I had better say what I have to say now. Better to have it out, before things go any further."

"Before *what* things go any further?" Phil asked, with a trace of belligerence. "Of course, if you don't want me for dinner—"

"Wait!" Katherine said in distress. "This isn't.... But it should be...." She looked from one to the other and smiled a tentative, hopeful little smile. "We don't have to go on with this, do we ... *now*?"

"What do you mean, 'now'?" Farley said, his face becoming red. "I think it's high time I got this off my chest. Katherine, I don't believe in letting things drift. I want this out in the open!"

(Oh, but this wasn't the way things were to be! This is all wrong—what can have happened?—There was no answer.)

Phil's face was pale and he started for the door. "I guess I'd better leave you two alone."

"No!" Farley said abruptly. "I want you here! I want you to hear this. Well, Katherine?" He turned to her again.

"I ... I can't answer you," she said miserably.

"You mean you are in love with him, don't you?" Farley said, with a kind of angry triumph. "All the time, behind my back, you—"

"Dry up, Farley!" Phil said, coming back from the door. "And stop acting like a bully!"

"Why, you—"

The telephone rang, and Katherine picked it up.

"It's for you, Russ," she said, and handed it to him.

"Yes?" Farley snapped into it. "Put him on." He listened for a few moments and his eyes traveled to Phil. "All right," he said. "When do you want to come?—I see. Well, I'll arrange to have him here. Three o'clock tomorrow,

then. Right. Good-by." He hung up. "You were right in one respect," he said to Phil. "That was a man from the Atomic Energy Commission and he wants to have a look at what you're doing. He'll be here tomorrow and I shall expect your full cooperation. Sorry, but it can't be helped."

Phil looked at him steadily for a moment. "So that's your way of getting back at me," he said. "Academic freedom means a lot to you! Of all the cowardly, spineless, rotten—"

Farley's face was now dark red and he held up his hand. "That's enough from you!"

"What the hell does he want to come nosing around here for?" Phil said. "My research is purely theoretical—"

"You yourself suggested they might," Farley reminded him. "And don't forget Lisa Meitner's work was theoretical, and look where it led!"

"Very cute!" Phil said. "In fact, Jesuitical! What I'm objecting to is having you dump me in their laps! I know your real motive—it stands out a mile!"

Farley's neck veins became noticeable, but he kept himself in control. "You tend to overrate your position here."

"Ha!" Phil said. "You can't bear me in my position of the man your wife loves!"

Farley's control went and he rushed at Phil and grabbed him by the collar.

"*Stop* it!" Katherine cried. "Stop it at once! Are you going to act like a pair of apemen? I'm not in love with Phil—I like him very, very much, but it's you I love, you ox!" She pushed and pulled, and they came apart like a bread sandwich, and she got between them. "For heaven's *sake*!"

"I'm sorry," Farley muttered, and looked ashamed of himself. "I wasn't *dumping* you into their laps, Kaufman—I had no choice. If I'd objected, they'd have got just that much tougher."

"It's okay," Phil said dispiritedly. "I guess."

"Oh, forget the whole thing, you two," Katherine said. "Come on, well be late for dinner." Taking their arms, she led them out onto the campus.

*** END OF THE PROJECT GUTENBERG EBOOK THE BACK OF
OUR HEADS ***

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